

Ensemble Learning

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Introduction

Ensemble Learning

- The *No Free Lunch Theorem* states that there is no single learning algorithm that in any domain always induces the most accurate learner
 - No single algorithm (model) is always the most accurate

Model	Bank Customer Churn Dataset	HR Analytics Job Change of Data Scientists	Drug Consumption	Internet Firewall
KNN	83.85 (3 ^o)	79.90 (1 ^o)	40.58 (1 ^o)	93.10 (1 ^o)
SVM	81.65 (4 ^o)	78.05 (4 ^o)	39.25 (3 ^o)	79.54 (4 ^o)
XGboost	86.97 (1 ^o)	79.37 (3 ^o)	38.58 (4 ^o)	92.98 (2 ^o)
Modern NCA (Ye et al., 2025)	85.42 (2 ^o)	79.58 (2 ^o)	39.89 (2 ^o)	92.92 (3 ^o)

Results available at: <https://6sy666.github.io/TALENT-Results/>

Ye et al. *Revisiting Nearest Neighbor for Tabular Data: A Deep Tabular Baseline Two Decades Later*. ICLR, 2025

Introduction

Ensemble Learning

- Ensemble learning is a paradigm that trains several low-accuracy models instead of trying to learn one super-accurate model
 - A collection of methods that learns a target function by training a number of individual learners and combining their predictions
- Popular ensemble learning algorithms
 - Bagging
 - AdaBoost
 - Random Forest

Weak Learners

Ensemble Learning

- Weak learners are learning algorithms that cannot learn complex models
 - Low-accuracy models
 - Often referred to as models performing slightly better than random guessing
- A popular weak learner is a form of decision tree known as a decision stump
 - Decision tree with a single node

Problem Definition

Ensemble Learning

- Assuming weak learners are deterministic algorithms (e.g., kNN, decision trees, logistic regression), how can we learn different weak learners?
- How do we combine the outputs of base learners for maximum predictive ability?

Bagging

Ensemble Learning

- The idea behind bagging is to generate K distinct training sets by sampling with replacement from the original training set
- The process is simple and consists of the following steps:
 - Randomly pick p samples from the training set
 - Train a **weak model** on these p samples
 - Repeat this process K times, generating K different weak models

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Subsampling of Features

Ensemble Learning

- Besides bagging, another way to generate K distinct weak learners is by training individual classifiers on random subsets of features (subspaces)
 - This scheme composes the basis for random forest
- The process is simple and consists of the following steps:
 - Randomly pick N features from the original set of m features
 - Train a weak model on these N features (subspace)
 - Repeat this process K times, generating K different weak models

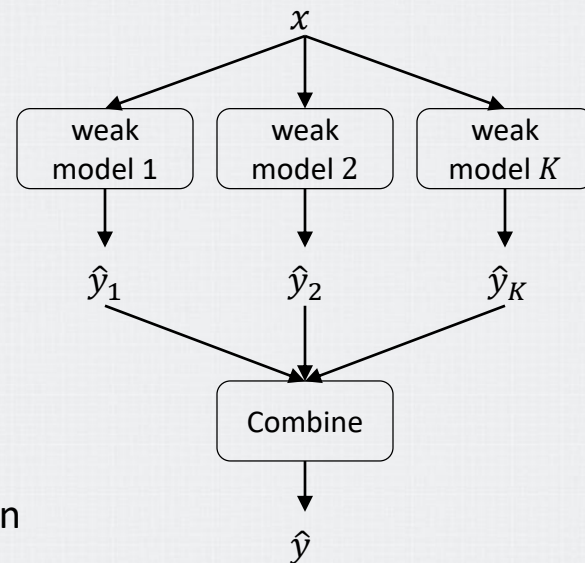
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Combining Predictions

Ensemble Learning

- Given k weak models and a new, unseen, sample x to predict, the most popular techniques for combining predictions are
 - Averaging (for regression)
 - Majority voting (for classification)
- Averaging
 - $\hat{y} = \frac{1}{k} \sum_{i=1}^k \hat{y}_i$
 - (weighted version) $\hat{y} = \frac{1}{k} \sum_{i=1}^k w_i \hat{y}_i, w_i \geq 0, \sum w_i$
- Majority voting
 - $\hat{y} = \operatorname{argmax}_{c \in C} \sum_{i=1}^k I(\hat{y}_i = c)$
 - C denotes the set of classes and I is the indicator function



Random Forest

Introduction

Random Forest

- Random forest is one of the most popular ensemble algorithms
- It creates an ensemble of decision trees using **bagging** and **subsampling of features**
 - Remember that decision trees are deterministic
- The reason for feature subsampling is to avoid correlation among the trees
 - If one or a few features yields a large impurity reduction, most trees repeatedly choose those features for splitting, thus, decreasing ensemble diversity
- An important parameter in random forest is the number of trees
 - Number of ensemble members

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Boosting

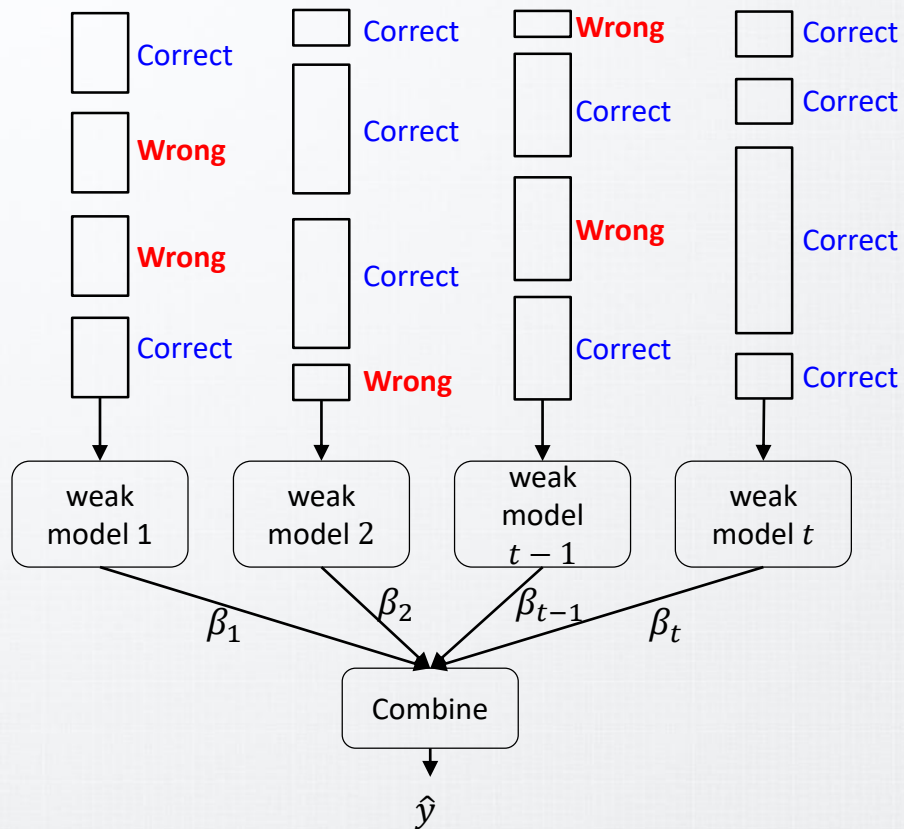
Introduction

Boosting

- The idea is to modify the probabilities of drawing the samples as a function of the error (weighted training set)
 - For example, if one example had a weight of 3 and the other examples all have a weight of 1, that would be equivalent to having 3 copies of the one example in the training set
- A weighted training set has the effect of focusing more on the samples the model is incorrectly classifying in the next round of boosting, and less on those it is correctly classifying
- Popular boosting algorithms
 - Adaboost
 - Gradient boosting

AdaBoost Algorithm

Boosting



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Sample weighting coefficient and Prediction

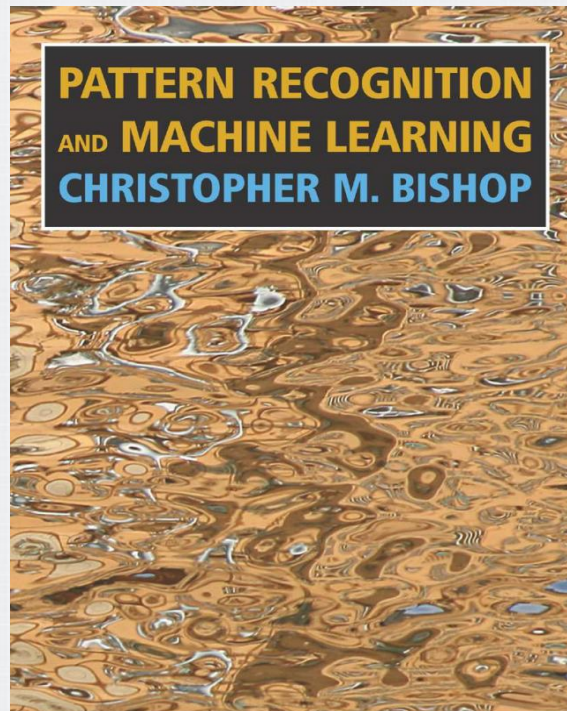
Boosting

Error Value ϵ	$\beta = \frac{\epsilon}{1 - \epsilon}$	Update correct sampling weights by β	$\beta = 0.5 * \log\left(\frac{1 - \epsilon}{\epsilon}\right)$	Update wrong sampling weights by $\exp(\beta)$
0.01	0.01	0.01	2.2976	9.9499
0.1	0.11	0.11	1.0986	3.00
0.2	0.25	0.25	0.6931	2.00
0.3	0.42	0.42	0.4285	1.5275
0.4	0.66	0.66	0.2027	1.2247
0.5	1.0	1.0	0	1.0
0.51	1.04	1.04	-0.20	0.9802
score model i	$\log(1/\beta)$		β	

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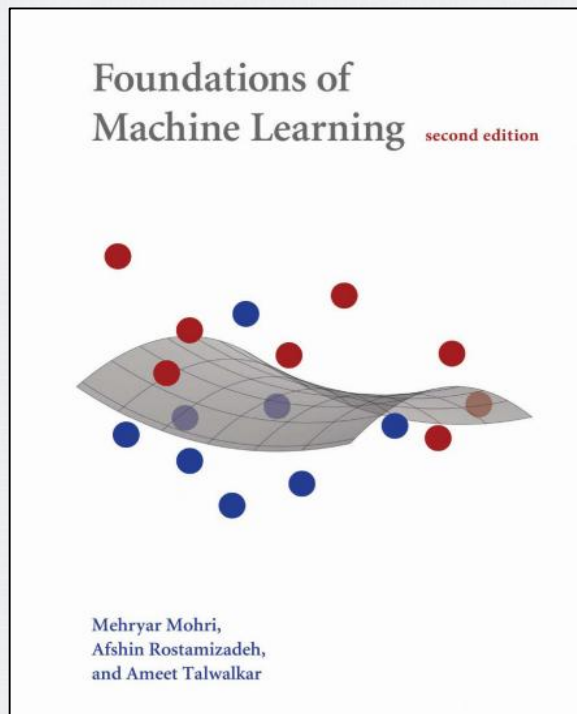
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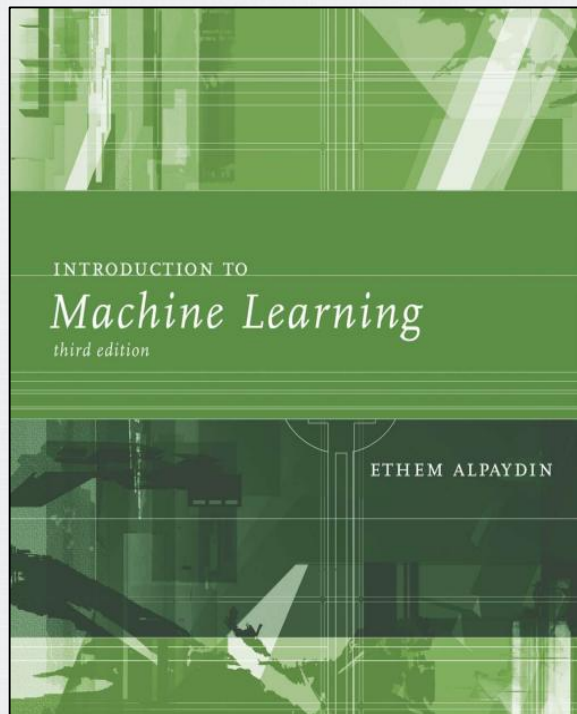
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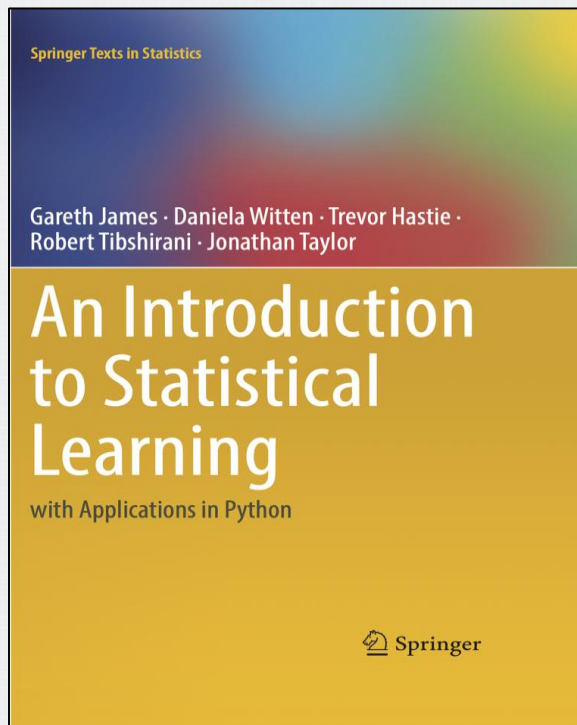
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